Henan Wang

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RESEARCH INTERESTS

Device–circuit co-design and DTCO for emerging memory and spintronic devices, spanning device physics, compact modeling, digital/analog/mixed-signal circuits, and BEOL-compatible monolithic CMOS+X integration.

EDUCATION

Purdue University, West Lafayette, IN Aug 2023 – May 2027
Bachelor of Science in Electrical Engineering *Concentration:* Microelectronics and Semiconductors
GPA: 3.88/4.00

- **Relevant Coursework:** Semiconductor Memory Technologies, CMOS Analog Design, Digital Design, Nanoelectronics, Microprocessor Systems and Interfacing, Electromagnetics, Structure and Properties of Materials
- **Honors/Awards:** Purdue Summer Undergraduate Research Fellowship (SURF) with Distinction, 2026
  Dean's List and Semester Honors, six semesters

PUBLICATIONS AND MANUSCRIPTS

Under Review

[1] Z. Lin*, **Henan Wang***, et al., "Realizing Dual-Mode Memristor Based on $Ni_{0.8}Fe_{0.2}$-BTO/BTO/ $Ni_{0.8}Fe_{0.2}$-BTO Tri-layer Vertically Aligned Nanocomposites," *Small*, under review after two rounds of revision.

[2] **Henan Wang***, Z. Lin*, et al., "Device–Algorithm Co-Design Enabling a CeO_2-based Vertically Aligned Nanocomposite Memristor for Neuromorphic Computing," *IEEE EDTM 2027*, under review.

**Co-first authors.*

In Preparation

[3] "Variability-Aware Device Modeling of Oxide-Based Vertically Aligned Nanocomposite Memristors."

Additional co-authored manuscripts: MTJ spin-dependent transport, MTJ-based p-bit circuits.

RESEARCH EXPERIENCE

Wang's Thin Film Group | Undergraduate Researcher Jan 2025 – Present
Purdue University, School of Electrical and Computer Engineering | *Research Assistant, Summer 2025*

Focus 1: RRAM Compact Modeling, Programming, and Neuromorphic Simulation

- Fabricate vertically aligned nanocomposite (VAN) RRAM via PLD and sputtering; correlate microstructure and interfaces with switching, multilevel and pulse-driven synaptic behavior using probe station, XRD, ellipsometry, AFM, and TEM; automate characterization and parameter extraction through MATLAB/Python workflows.
- Developed a measurement-calibrated behavioral compact model and model-informed program-and-verify control; experimentally improved 10 s settled-state targeting from 4/18 to 17/18 within ± 0.05 decade, reducing pulses per success from 14.8 to 2.9. Simulations using measured programming errors achieved 95.9% MNIST accuracy and 12$\times$ lower GPT-2 perplexity versus fixed-amplitude programming. [2]
- Built a device aware crossbar simulation for the dual mode memristor, extracting update nonlinearity, polarity reversal steps, C2C/D2D variation, read/write noise, and retention from raw pulse data; with write-verified differential-pair synapses, reached 92.7% MNIST accuracy (96.2% software) . [1]

Focus 2: Thin-Film Heterostructures and Device Physics

- Co-develop thin film material systems, growth conditions, and mechanism hypotheses for ferroelectric/ferromagnetic, VO_2 phase change, and 2D-barrier devices; investigate how heteroepitaxy, strain, interfaces, defects, and nanocomposite morphology govern ferroic, magnetic, optical, and electronic properties.
- Investigate transport and switching physics in multifunctional magnetic tunnel devices, including spin-dependent tunneling, nonvolatile-state retention, stochastic switching, and optical control of stochasticity; correlate material structure and device transport to identify mechanisms underlying deterministic and probabilistic operation.

NanoX Lab | Undergraduate Researcher Jun 2026 – Present
Purdue University, Birck Nanotechnology Center | *SURF with Distinction, Summer 2026*

Focus 1: BEOL Integration and Automated Characterization of CMOS+X (Emerging-Device) Systems

- Contribute to MTJ-array development and study BEOL process planning and CMOS compatibility through a polar-capacitor majority-logic project.
- Pursue cleanroom training and qualification in lithography, thin-film deposition, and etching for CMOS+X integration.
- Built an NI PXIe platform combining multichannel SMUs, digital-pattern instruments, and FPGA resources to test emerging devices and CMOS+X prototypes.

Focus 2: MTJ p-Bit Reversible Full-Adder for Stochastic Computing (in production at TSMC)

- Contributed to RTL design, debugging, and mixed-signal integration of an MTJ p-bit based reversible full-adder for stochastic and probabilistic computing.
- Designed, simulated, and laid out MUX and inverter circuits for MTJ output selection and signal restoration; developed Verilog-A stochastic device models and RTL-compatible analog-macro interfaces.

Circuit & Hardware Design Projects

Senior Design – Cryogenic Bias-Voltage DAC Aug 2026 – Present

- Developing a programmable bias-voltage DAC for cryogenic applications in Cadence Virtuoso; tape-out planned for Spring 2027.
- Implemented and simulated a MOS reset in a 3-bit TSMC 65 nm MIM capacitor DAC with six transmission-gate reference switches; compared reset sizing and charged-output discharge.
- Observed a 179.9 μV code-111 output spread in finite-delay TT simulations at $-40/27/85\,^{\circ}$C with ideal 1 V references and no added load; cryogenic validation remains pending.

Fixed-Point Digital Attention Accelerator *Chips and AI Hackathon (Solo)* Jul 2026 – Present

- Designed a synthesizable signed-int8 scaled dot-product attention accelerator spanning MAC, dot-product, matrix multiplication, systolic-array, QK^T, LUT-based exponential, fixed-point normalization, and PV datapaths.
- Replaced 16 sequential divisions with four row-wise reciprocals and multiply-and-round normalization, reducing the compute window from 577 to 205 cycles. Matched a bit-accurate Python model on the same 216 input vectors in RTL and post-layout gate-level simulation.
- Removed a 48-cycle serial feed with parallel operand delivery; overlapped host I/O using banked input and output storage, reducing the task interval from 329 to 213 cycles. Completed RTL-to-GDSII implementation using a commercial 40 nm CMOS PDK, with setup/hold slack of +0.046/+0.030 ns at 100 MHz.

Precision Bandgap Voltage Reference *AMS Team Member* Jun 2025 – Dec 2025

- Independently designed and simulated an approximately 1.5 V CMOS bandgap reference in Cadence Virtuoso/Spectre, combining PTAT and CTAT components and optimizing device/resistor ratios for first-order temperature compensation.
- Achieved 5.7 ppm/°C nominal TC over -40 to 125 °C, $<$ 1% Monte Carlo output variation, and 60 dB PSRR at 100 kHz.

SoCET – AFTx07 (Taped-Out): DFT *Test Engineering Team Member* Jan 2025 – May 2025

- Customized a Synopsys Fusion Compiler flow for UART block RTL synthesis, timing/area constraints, optimization, technology mapping, and gate-level netlist export.
- Inserted 4 scan chains with zero DFT DRC violations and developed stuck-at ATPG; the team generated 62 scan patterns and achieved 99.18% stuck-at test coverage on the UART block.

SELECTED PRESENTATIONS

Henan Wang, "Design and Modeling of a Bias Voltage DAC for Cryogenic Qubit Control," *2026 Fall Undergraduate Research Expo*, Purdue University. **Selected for virtual presentation.**

Henan Wang, "Design of an Automated Characterization Infrastructure for CMOS+X Integrated Systems," *Purdue Undergraduate Research Conference: 2026 Summer Symposium*, West Lafayette, IN, Jul. 2026. **Poster presentation.**

Henan Wang, "Design and Characterization of a Precision Bandgap Voltage Reference for CMOS Analog Systems," *Purdue Undergraduate Research Conference: 2025 Fall Expo*, West Lafayette, IN, Nov. 2025. **Oral presentation.**

TEACHING EXPERIENCE

ECE 20001 – Electrical Engineering Fundamentals | Undergraduate Teaching Assistant 3 Semesters

- Led recitation sessions when the instructor was attending conferences or was otherwise unavailable.
- Held office hours and answered questions on circuit analysis and electrical engineering fundamentals on Piazza.
- Proctored examinations and graded assignments, quizzes, and exams.

SKILLS

Device Fabrication: PLD, sputtering, photoresist spin coating, maskless direct-write photolithography (Heidelberg MLA150), wet/dry etching

Device Characterization & Test Automation: XRD, ellipsometry, AFM, TEM sample preparation, probe station, PMU/SMU, oscilloscope, NI PXIe, VISA-based instrumentation, modular DC/pulse measurement workflows, automated data logging and analysis

IC / Hardware Design: Cadence Virtuoso/Spectre, Cadence Genus/Innovus/Tempus, Synopsys Fusion Compiler/TestMAX, Verilog/SystemVerilog, Verilog-A, Altium Designer, EasyEDA

Programming & Data Analysis: Python, C/C++, MATLAB, Origin, SPICE, MNSIM